1820

> **“** THE **EXPERIMENTAL INVESTIGATION** BY WHICH **AMPÈRE ESTABLISHED** THE LAW OF THE **MECHANICAL ACTION** BETWEEN **ELECTRIC CURRENTS** IS ONE OF THE MOST **BRILLIANT ACHIEVEMENTS** IN SCIENCE. **”**

James Clerk Maxwell, British theoretical physicist, from *A Treatise on Electricity and Magnetism*, 1873

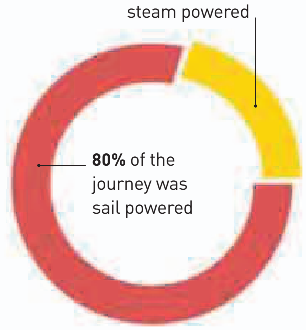
112°51’W

THE **LONGITUDE** IN THE **NORTHWEST PASSAGE** REACHED BY **HMS *HECLA*** AND **HMS *GRIPER***

weights proved to be a difficult task. Petit and Dulong found that an element’s **specific heat** (see 1761–62)—the amount of heat required to raise the temperature by one degree Celsius—is **inversely proportional** to its **atomic weight**. By measuring the specific heat of an element, Petit and Dulong were able to make an estimate of its atomic weight.

During this period, finding a sailing route to the Pacific through the Arctic was an attractive proposition for commercial interests in Europe, because the routes to the south were long and stormy. British naval officer William Parry led an expedition to find the **Northwest Passage** in 1819 and succeeded. He reached Melville Island in the Arctic and won the prize offered by Parliament for crossing a longitude of 110° West. Parry’s ships, HMS *Hecla* and HMS *Griper*, were trapped by frozen sea until the spring of 1820 when the ice finally melted.

Another ship, the SS *Savannah,* became the **first ship to cross the Atlantic using steam engines**, sailing from Savannah in the USA on May 22 and arriving in Liverpool in England 18 days later. This feat was not repeated for over 20 years. Despite being equipped with luxury cabins, however, no passengers could be persuaded to join the voyage because of the ship’s revolutionary design: a sailing ship that also contained a steam engine operated by paddle wheels.



Voyage of the SS *Savannah*
While the SS Savannah *was the first ship to cross the Atlantic using steam power, the ship used its steam engines for only 41½ hours during the 207-hour voyage.*

DANISH PHYSICIST HANS CHRISTIAN ØRSTED (1777–1851) revealed the **link between electricity and magnetism**. At a public lecture in Copenhagen, he astonished the audience by showing a compass needle move as he brought it near a wire conducting electricity. Influenced by Ørsted’s discovery, French physicist **André-Marie Ampère** created a theory of **electromagnetism**. This showed that electric currents flowing in opposite directions create magnetic fields that cause the wires to be attracted, while currents flowing the same way lead to the wires being repelled.

British physicist **John Herapath** (1790–1868) explained how temperature and pressure in gas are created by moving molecules, an early version of the **kinetic theory of gases**.

ANDRÉ-MARIE AMPÈRE (1775–1836)

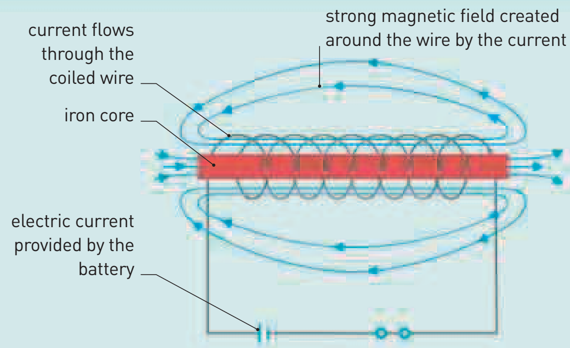
Born near Lyon in France, André-Marie Ampère was a talented mathematician and teacher. He laid the foundations of electromagnetic theory and discovered that the magnetic interaction of two electrical wires is proportional to their length and the strength of the current flowing through them. This is known as Ampère’s Law.

In Paris, French naturalist **Georges Cuvier** ridiculed the ideas of fellow naturalist **Jean-Baptiste Lamarck**, who argued that **species have transformed, or evolved**, through time. This idea is now an accepted part of the theory of evolution.

Also in Paris, chemists **Pierre-Joseph Pelletier** (1788–1842) and **Joseph Bienaimé Caventou** (1795–1877) worked on **isolating medically active ingredients from plants**. In 1820, they isolated quinine from cinchona bark, which later became important in treating malaria.

ELECTROMAGNETS

An electric current creates its own magnetic field, and this is the basis of an electromagnet. These are powerful magnets that can be switched on and off with an electric current. A solenoid (coil of wire) is a common form of electromagnet. The more coils of wire, the stronger the magnetic field. Electromagnets are vital to the operation of everything from the speaker in a phone to electric motors.



June 20 SS Savannah is the first **steamship** to cross the Atlantic Ocean

August British Arctic expedition under William Parry reaches longitude 112°51’ W

Captain John H. Hall developed the M1819 rifle using interchangeable parts

French chemists Pierre Dulong and Alexis Petit find an element’s atomic weight is **inversely proportional** to its specific heat

January 28 Sighting of Antarctica by Russians under Fabien Bellingshausen

January 30 Sighting of Antarctica by British under Edward Bransfield

April 21 Danish physicist Hans Christian Ørsted shows the **link between electricity and magnetism**

July Hans Christian Ørsted publishes a pamphlet setting out his ideas on **electromagnetism**

September 20 André-Marie Ampère begins work to establish the basic theory of **electromagnetism**

November 17 Sighting of Antarctica by American Nathaniel Palmer